

Comparison of Structure Response to Hard and Soft Rock Blasting

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Abstract

The response of two structures to blast-induced ground vibrations were evaluated and compared in order to quantify the impact of different blasting operations, hard rock and soft rock blasting, upon the structures movements. The first instrumented structure was a residential type, located in the operational area of a coal mine, where blasting is conducted in soft rocks, such as siltstone and coal, with compressive strength ranging from 20 to 50 MPa. Blasting is characterized by low benches, with hole lengths averaging 5.5 m (18 ft) and low powder factors, 140 to 210 g/m³ (0.236 to 0.354 lb/yd³), resulting in very little horizontal movement of the rock during the blast. The second structure was a commercial building, located in a basalt quarry (compressive strength between 109 and 180 MPa), where blasting is conducted in 12 m (39.4 ft) benches with powder factors ranging from 200 to 350 g/m³ (0.337 to 0.590 lb/yd³). Velocity sensors were placed on the structure walls to measure wall displacements and calculate blast-induced strains in the walls, and these results were compared with the driving ground excitations. The calculated strains were compared with the limits of rupture of the cement grout used as mortar between bricks and as wall coating, considered to be the weakest material in the structure composition. The blast-induced response of existing cracks in the structures were also measured using displacement gages and compared to the daily influence of weather fluctuations, such as changes in temperature and humidity, on crack movement. The results show that for the same levels of ground motions the structures present different movement, resulting in distinct levels of induced strains on the walls depending on the geology. The influence of long-term temperature and humidity variations on crack movements on both structures is very similar, during the same seasons of the year. Crack response to long-term weather changes is 2.5 to 3.5 times greater than blast influences.

Introduction

Aiming to evaluate the impact of blasting vibrations on structures, a series of structure response studies are being conducted in common Brazilian structures at different blasting operations. This paper presents a comparison of the results of the first two studies conducted in a coal mine and a quarry. The objective of this comparison is to show the influence of the geology, and, by consequence, the type of blast designs on the structures response.

Figure 1 show pictures of the two structures selected for the studies and are examples of typical constructions found in the vicinity of mining operations all over Brazil. Typical Brazilian structure walls are built of masonry clay bricks with six holes and cement mortar. Foundations comprise concrete beam perimeter walls and thin concrete slabs. Roofs are very heavy as they are construction with clay tiles over a wood frame. Typically, walls are covered with cement grout, spackling (or plastered) and then painted, and oftentimes no type of cover is used in the bricks walls.



Figure 1: Structures instrumented at the coal mine (left) and basalt quarry (right).

The study conducted at the coal mine lasted for approximately six months, between August 2012 and February 2013, and 148 blasts were conducted during this period at distances varying from 32 to 810 m (105 to 2,657.5 ft). A total of 115 blasts were recorded at the structure and from these, 104 were evaluated. Eleven events were disregarded because of a failure of the slope excavated near the structure, 5 m (16.4 ft) away, which resulted in foundation movement impairing the recorded data. The study at the quarry started in March 2013 and ended in April 2014, and 14 blasts were recorded during this period, at distances between 86 to 300 m (282 to 984 ft). Although the number of events is much smaller the results are considered representative.

Blasting Site Characteristics

Coal mine blasting is characterized by the lower powder factor used, 140 to 210 g/m³ (0.236 to 0.354 lb/yd³), as the geology is composed by soft rocks (siltstone and coal), which do not require much energy to be disaggregated. These types of rocks have a compressive strength ranging from 20 to 50 MPa (Goulart, 2002). The use of low powder factor leads to very low horizontal movement of the material and can result in high levels of vibration (Rosenhaim *et. al.*, 2012). Other important parameters of the design are blast hole length or bench height and drill patterns for each material. During the study, hole lengths varied from 2.6 m (8.5 ft) to 9.1 m (30 ft), averaging 5.9 m (19.4 ft), in 3 and 3.5 in (76.2 and 89 mm) holes. Drilling patterns varied from 3 × 4 m (9.8 × 13.1 ft) in coal shots to 4 × 5 m (13.1 × 16.4 ft) in overburden blasts, with 3 in holes, and 4.5 × 6.5 m (14.7 × 21.3 ft) in overburden blasts, with 3.5 in (89 mm) holes.

Powder factors used in quarry blasting are higher than in coal blasting. Hard rocks, such as granites and basalts, require more energy to be moved and fragmented. Usually, bench heights are over 10 m (33 ft) and drill patterns are much tighter. At the basalt quarry, powder factors ranged between 200 and 350 g/m³ (0.337 and 0.590 lb/yd³) and bench height was 12 m (40 ft). The drill pattern for most blasts was 2.5 × 4.5 m (8.2 × 14.7 ft) in a 2.5 in (63.5 mm) diameter hole. The compressive strength of the basalts extracted at the quarry ranged from 109 to 185 MPa (CIENTEC, 2014).

Table 1 compares maximum and minimum values for some parameters of the blast designs and results of ground vibration and air sound pressure amplitude (or airblast) measured at both sites. The data show that besides the difference in powder factor used in each blast design, the variation in average peak frequency and predominant frequency (FFT) are significant. Blasting in the coal mine (soft rock) resulted in ground motions with low frequencies, while at the quarry (hard rock) the resulting vibrations generate higher frequencies.

Table 1: Blasting design parameters and resultant ground vibrations and air overpressure ranges

Blast Site	Powder Factor (g/m ³)	Charge Weight/ Delay (kg)	Distance (m)	PPV (mm/s)	Peak Freq. (Hz)	FFT (Hz)	Air Sound Pressure Level (dB)
Coal Mine	140 – 210	18.2 – 245.9	57 – 660	1.27 – 36.57	8 – 15	12.6	100 – 148
Quarry	200 – 350	24.0 – 120.7	86 – 290	3.56 – 20.83	20.2	21.3	120 - 144

Figure 2 (a) presents the Brazilian Standards for safe blasting in urban areas (ABNT NBR 9653, 2005), where peak particle velocity is plotted against peak frequency for all data recorded in the ground at the studied structures. The levels of ground vibration recorded over the proposed limit at the structure in the coal mine did not result in any observed damage to the structure. In the attenuation plot presented in Figure 2 (b), peak particle velocity is plotted with scaled distance for each event recorded at the structures and with linear arrays of seismographs as well as other monitoring points.

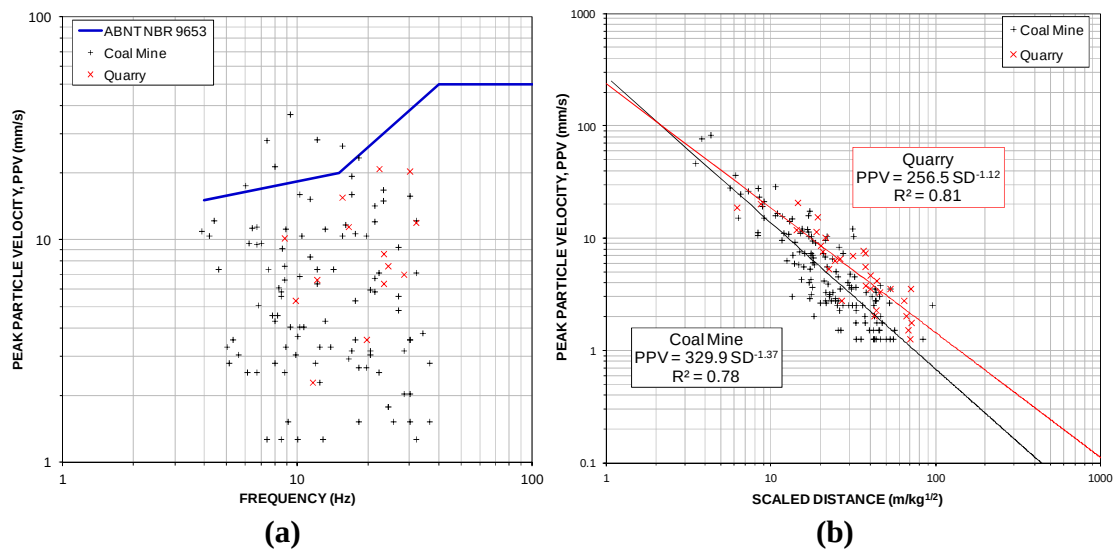


Figure 2: Brazilian Standard for safe blasting in urban areas (a) and ground vibration attenuation plot (b).

Comparing the attenuation equations for both blasting sites, the higher y-intercept shown by the equation for the coal mine indicates that the use of lower powder factors in the blasting of soft rocks results in higher confinement and higher close-in amplitudes of vibrations. On the other hand, the differences in the slopes of the lines indicates that the high levels of vibration recorded close-in at the coal mine dissipate quicker than for the quarry. At the quarry, higher intensities of ground motions are carried farther away from the blast and present a slower decay rate.

Measuring Structure Movement

In order to measure whole structure movement and compute wall strains, single axis geophones were mounted in the structures upper (S2) and lower (S1) corners and mid walls (MW), as shown in Figure 3. Structure movements were time-correlated with ground vibrations by an exterior geophone buried 1 m (3 ft) from the instrumented wall corner.

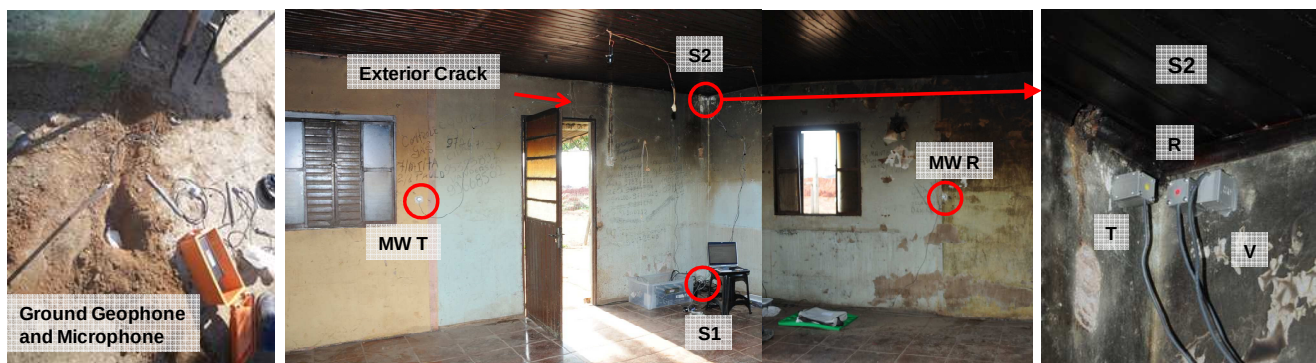


Figure 3: Exterior ground geophone and microphone (left) and single axis geophones placed on the coal mine structure walls (center and right).

The plots presented in Figure 4 show correlations of the lower structure, upper structure and mid-wall movements with ground vibrations (in terms of velocity). From the comparison of lower corner with ground motions it is possible to see that both structures are well coupled with the ground, resulting in a small decrease in the vibration amplitudes at the foundation. The upper corner movement of the coal mine structure shows a small amplification of ground vibrations compared to the response demonstrated by the upper structure movement of the quarry structure, especially in the radial direction, facing the blasts. Mid-wall response to ground movement is more significant in the quarry structure.

Natural Frequency

The structures were determined to be very rigid based on construction and did not demonstrate any free response after the cessation of the ground excitation. The structures moved together with the ground and after the passing of the seismic waves stopped moving. For this reason, in order to evaluate the natural frequency of the structures a technique involving transfer functions calculated from Fourier frequency analysis was used (Dowding, 1985). The method consists of applying the Fourier transform integral to the velocity time-history waveform to yield the power spectral density function. The ratio between the square of the Fourier amplitude of the upper structure (S2) and ground (GV) was obtained and compared to the square of the spectral density of the upper and ground response to determine the maximum structural response.

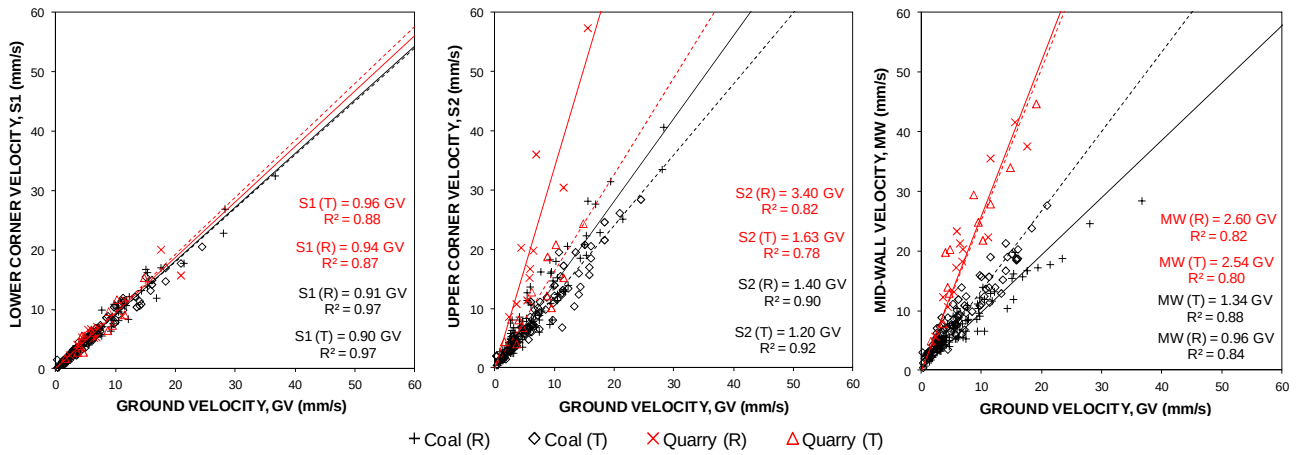


Figure 4: Comparison of the response of the structures to ground vibrations.

The frequency analysis resulted in a natural frequency range from 7.5 Hz to 12 Hz, depending on the axis of the structure. For both structures, the wall parallel with the longest axis of the structure and perpendicular to the roof frame showed the highest natural frequency, ranging from 10.5 to 12 Hz. The walls perpendicular to the longest axis of the structure and parallel with the roof frame showed a lower natural frequency, from 7.5 to 9 Hz. These results indicate the influence the heavy roof frame exerts on the structure movement.

Computing Blast-Induced Strains

Velocity measurements recorded by the seismographs were integrated to determine displacements and the differential wall displacements between upper and lower corner movements were computed ($S2 - S1$). The maximum differential wall displacement (δ_{max}) was then used to determine the maximum induced shear strain (γ_{max}) in the wall (Dowding, 2000). Figure 5 (a) presents a schematic drawing of the geometry of the wall during shear strain and the equation used for calculations. In-plane tensile strains (ϵ_{Lmax}) is the deformation mode most likely to cause cosmetic cracks in walls during high levels of structure motion and is directly related to global shear strains in the walls (Aimone-Martin *et. al.*, 2003). The equation and geometry of the wall used to compute in-plane tensile strains are shown in Figure 5 (b). Mid-wall bending strains are directly related to bending stresses induced in the walls, and are estimated using the equation and wall geometry presented in Figure 5 (c). In order to determine the true mid-wall displacement, it is necessary to consider the movement of the actual mid-wall in relation to the whole wall movement as determined by the average between upper and lower structure displacement. The true mid-wall movement is then computed as the measured mid-wall displacement subtracted from the position of the wall during this displacement.

Calculated strains were compared with the failure limits of each material used in the construction, and the probability of causing superficial cracks in the walls was estimated (Aimone-Martin *et. al.*, 2003). The cement mortar used between bricks and as wall coating (interior and exterior walls) was considered the weakest material in the structures and, therefore, the failure limits of this material were used to evaluate crack potential. Studies conducted in cement mortar indicate the failure limits of this material by shear strain ranges between 1,400 and 1,600 micro-strains and by in-plane tensile and mid-wall bending strains from 150 to 250 micro-strains (Paes and Carasek, 2002, Mohamad *et. al.*, 2009). Table 2 summarizes the ground vibrations and the resulting maximum differential displacement, shear strain and

in-plane tensile strain for each structure. The PPV which drives the maximum mid-wall bending strains is also presented. It is noticed that for the coal mine structure, even with a higher ground vibration, the levels of strain produced in the walls are smaller than the resultant strains computed for the quarry structure. Maximum mid-wall bending strains are similar for both structures.

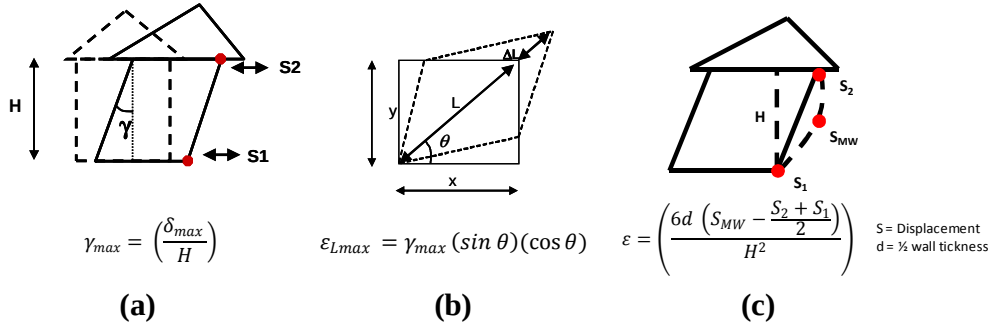


Figure 5: Geometry of the wall and equations used to compute shear strains (a), in-plane tensile strains (b) and mid wall bending strains (c) (modified from Rosenhaim *et. al.*, 2005).

Table 2: Summary of maximum computed strain levels and driving ground vibrations

Blast Site	Ground Vibration (mm/s)	Peak Freq. (Hz)	Differential Displacement (mm)	Shear Strain (μ -strains)	In-plane Tensile Strain (μ -strains)	Ground Vibration (mm/s)	Peak Freq. (Hz)	Bending Strains (μ -strains)
Coal Mine	28.19	12.1	0.331	132.2	39.2	14.99	8.1	33.4
Quarry	15.49	15.5	1.190	475.8	237.2	14.73	22.2	37.3

The blast at the coal mine resulting in the highest strain level (132.2 μ -strains) was 57 m (187 ft) away from the structures with a charge weight per delay of 102.1 kg (225 lb). This resulted in a scaled distance of 5.6 m/kg^{1/2} and a PPV of 28.2 mm/s (1.11 in/s). In contrast, the quarry blast with the highest strain was 140 m (459.3 ft) from the structure with a charge weight per delay of 52.9 kg (116.6 lb) and resultant scaled distance of 19.2 m/kg^{1/2} and PPV of 15.5 mm/s (0.61 in/s). The resulting strain level was 475 μ -strains. This shows that even though the blast at the quarry resulted in lower PPV with a higher scaled distance (lower charge weight per delay and greater distance from the structure), blasting in hard rock resulted in a higher strain amplitude compared with the coal mine site structure, for the same levels of ground vibrations. Also, it must be considered that the structure at the quarry was of different shape and size compared with the structure at the coal mine, with shorter walls, which allowed more movement with the passing of the ground motions.

Figure 6 shows correlation plots of the induced strains with the driving ground motions and compares the two blast sites. The results shows higher levels of induced wall strains for the quarry structure at the same levels of ground vibrations recorded at the coal structure. This indicates that for the same levels of vibration, the response of this structure in hard rock is more sensitive than the structure in soft rock.

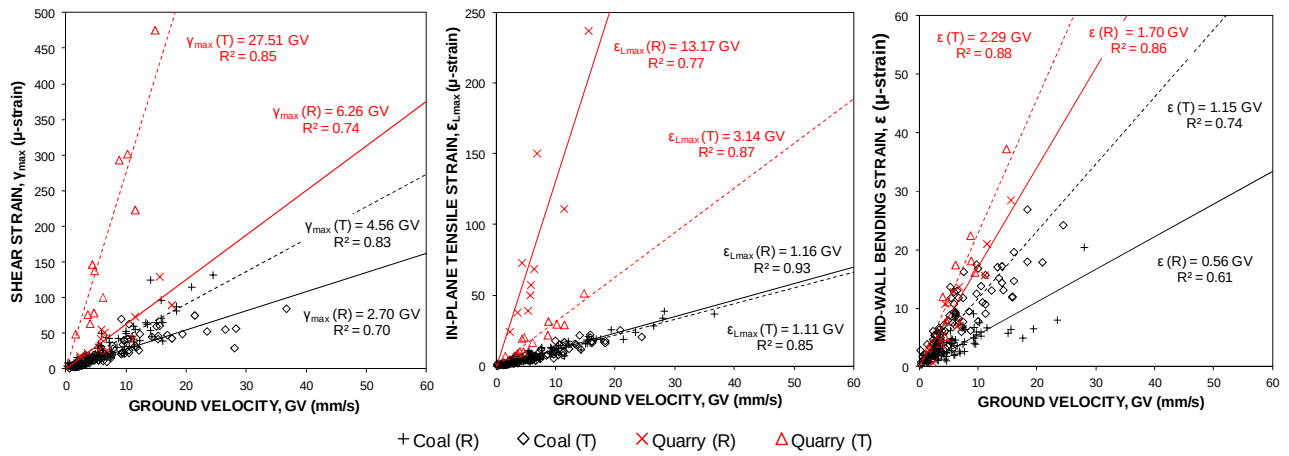


Figure 6: Correlation plots of computed induced wall strains with driving ground vibrations.

Comparing the maximum strains presented in Table 2 with the failure limits of the cement mortar, it was possible to calculate factors of safety (FOS) against cracking, as shown in Table 3. All FOS values are well above 2.0 with the exception of those for quarry blasts. Even though the values for quarry blasting are at or less than 1.0, no new cracks, as well as the extension or opening of existing cracks in the structure monitored at the quarry were observed. The levels of particle velocity produced by the blast resulting in the highest levels of deformation (structure deformation) were not the largest recorded at the structure. However, peak and predominant FFT frequencies were below the average of the other events recorded. PPV for the blast resulting in the highest strains was 15.5 mm/s (0.61 in/s) with a peak frequency of 15.5 Hz and FFT frequency of 7 Hz. The average peak frequency for all other events was 20.6 Hz and average FFT was 24.2 Hz. The FFT frequency in the ground of the blast with the highest strain at the quarry was near the natural frequency of the structure in the same direction (8.2 Hz). Ground displacement for this blast was 0.220 mm (0.00866 in), while for the blast with highest PPV (20.83 mm/s, 0.82 in/s) peak displacement was 0.160 mm (0.00623 in). These results show that frequency exerts great influence over ground displacement, as ground displacement is inversely proportional to ground frequency ($Disp = PPV/2\pi f$), the lower the frequency in the ground the higher the displacement. The structure movement is a function of ground displacement and frequency and, as a consequence, the lower the frequency in the ground, closest is the match to the natural frequency of the structure, and highest is the ground displacement, increasing the likelihood of high strains in the structure. Figure 7 shows comparison plots of differential wall displacements with ground displacement and in-plane tensile strains. This plots show that when ground displacement increases, differential wall displacement also increases and higher is the strain levels.

Table 3: Computed factors of safety against cracking

Blast Site	Factors of Safety against Cracking		
	Shear Strain	In-Plane Tensile Strain	Bending Strain
Coal Mine	10.6 – 12.1	3.8 – 6.3	4.5 – 7.5
Quarry	2.9 – 3.4	0.6 – 1.05	4.0 – 6.7

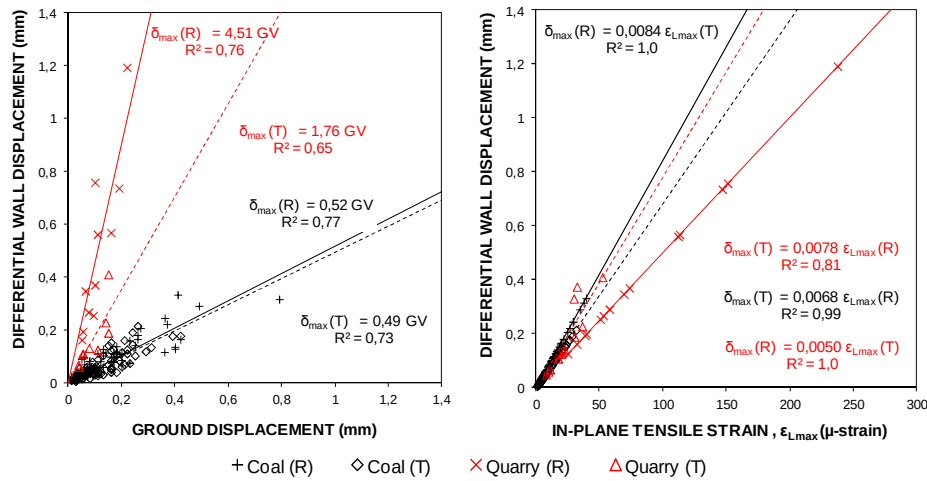


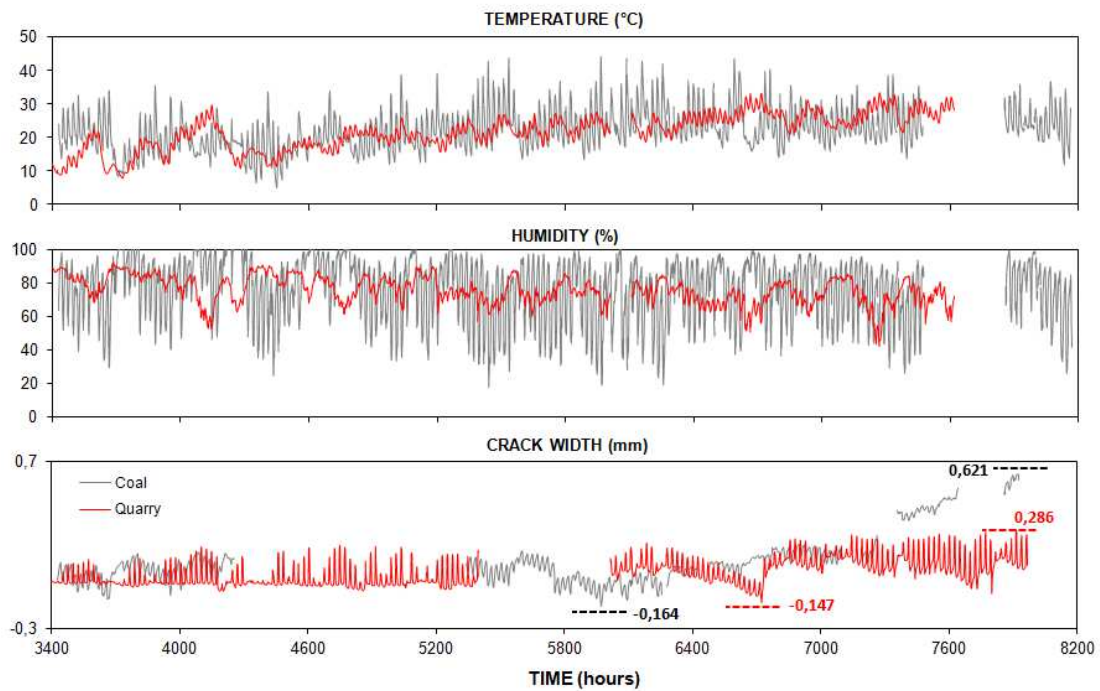
Figure 7: Correlation plots of differential displacement with ground displacement and in-plane tensile strains.

Differences in the computed blast-induced strains in the structures at the same amplitude of ground vibrations can be associated with the geology and mainly to the energy delivered by the explosives. As hard rock needs more energy to break, higher strain levels are expected. In this work, the variation in powder factor at the same blast site and the impact in the strain levels were not evaluated.

Crack Response

Existing cracks in the studied structures were instrumented with displacement type gages to evaluate the dynamic response of the cracks with the influence of blasting vibrations and this response was compared with the crack opening and closing resulting from the daily variation in temperature and humidity. Weather changes, especially daily variations in temperature and humidity, show great influence on the materials composing the structure walls. These forces begin at the time of construction and continue to exert high forces on construction materials during the entire life of the structure (Oriard, 1990). In order to show the weather-induced effects in the crack width, measurements of an existing crack aperture were taken on a hourly basis, 24 hours per day, for long periods of time (from 6 to 12 months). The variations in crack width are compared with daily changes in temperature and humidity (Figure 8) for measurements conducted at the coal mine and quarry structures for the same seasons of the year during each study period (2012/2013 at the coal mine, and 2013/2014 at the quarry). The periods compared range from August to February comprising the end of the winter, full spring and 2/3 of the summer, corresponding to 200 days of measurements.

It is observed in Figure 8 that with the increase in temperature during the approach of the summer, humidity decreased, especially during the day, and the crack aperture tended to open (showing a positive increase in the crack width). During daily cycles (12 hours) of weather changes, humidity shows more influence over crack movement than temperature. The higher the variation of humidity during one day, the higher was the crack movement (opening). The results shown in the table below Figure 8 indicate that when the variation in humidity is maximum, the crack width is higher than when temperature change is maximum.



Maximum Variation	Overall		Crack Width		Temperature		Humidity	
	Coal	Quarry	Coal	Quarry	Coal	Quarry	Coal	Quarry
Crack Width (mm)	0,785	0,434	0,181	0,132	0,133	0,115	0,148	0,187
(micro-in)	30915	17079	7126	5197	5236	4528	5827	7362
Temperature (°C)	39,2	27,3	15,1	-	27,7	8,5	26,1	3,6
Humidity (%)	82,3	52,9	50,9	-	65,4	13,1	69,2	27,9

Figure 8: Comparison of variations in temperature and humidity with crack width for the studied structures, during the same period of a year, end of the winter to mid to end of summer (200 days).

The overall variation in crack width, or the difference between the maximum opening and closure of the crack width, for the coal mine structure was 0.785 mm (30,915 micro-inches), while for the structure at the quarry the overall crack width was 0.434 mm (17,079 micro-inches). The maximum daily variation in crack width was 0.181 mm (7,126 micro-inches) and 0.319 mm (12,571 micro-inches) for the structure at the coal mine and quarry, respectively. Problems with the temperature and humidity sensor prevented recording the changes in temperature and humidity during the period were the daily variation in crack width was maximum. The second largest crack width variation was 0.265 mm (10,433 micro-inches) at a temperature variation of 4.5 °C and humidity change of 3.6% inside the building.

Figure 9 shows that as the variation in temperature increases, humidity decreases and crack width closes. The opposite occurs when temperature variation decreases and humidity variation increases, resulting in crack aperture opening. Temperature measurements in the quarry were taken inside the structure, where the crack was located, while in the coal mine, measurements were in the exterior of the structure. Nonetheless, the plots in Figure 9 show weather influence in crack width variation is more significant at the quarry structure.

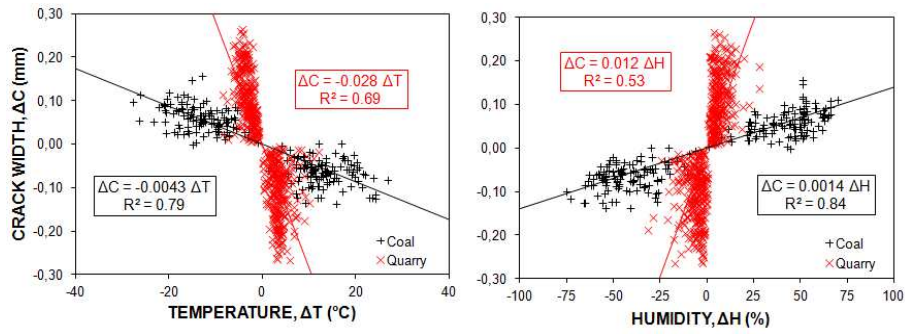


Figure 9: Correlation of variations in crack width with temperature and humidity daily changes (12 hr cycles).

Figure 10 (a) show the correlation of blast-induced dynamic crack response, in each structure, with ground vibration levels registered at the ground. The crack response in the coal mine structure is higher than for the quarry structure, given the same levels of vibration. The maximum crack width for the coal structure at a PPV of 14.99 mm/s (0.59 in/s) with a peak frequency of 23.2 Hz was 0.309 mm (12,165 micro-inches). For the quarry structure, the maximum crack width was 0.132 mm (5197 micro-inches) during a blast with PPV of 20.32 mm/s (0.8 in/s) and frequency of 30.1 Hz. Figure 10 (b) presents a comparison of dynamic crack response with computed in-plane tensile strain levels. The higher the strain level in the walls, resultant from higher ground motions, greater is the response of the existing cracks.

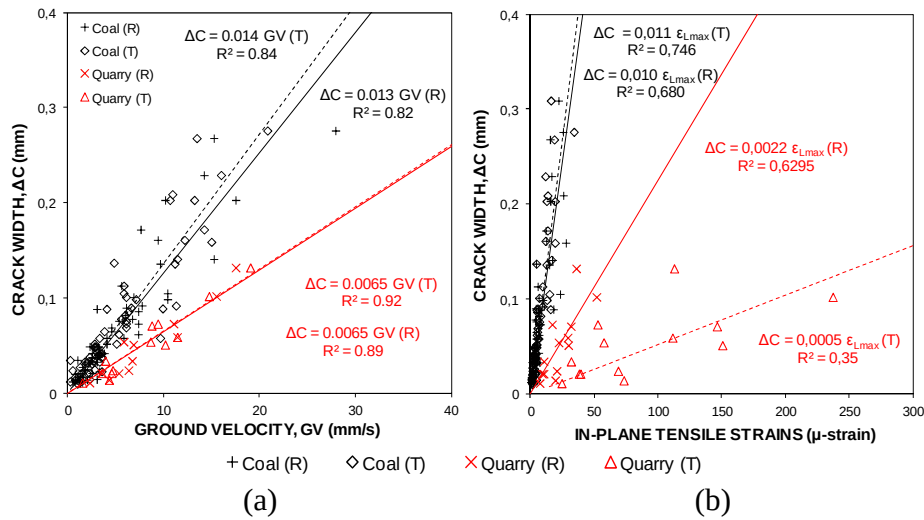


Figure 10: Correlation of plots of variations in crack width with ground velocity (a) and computed in-plane tensile strains (b).

Comparing the long-term weather-induced crack response with the maximum blast-induced crack movement, in Figure 11, the influences of temperature and humidity variations over time on the crack width are 2.5 to 3.5 times greater than blast influences at the quarry and coal mine, respectively. Therefore, it can be concluded that the variation in crack width resulting from weather oscillations is the leading factor contributing to the extension, and widening of existing cracks as well as the formation of new cracks over time.

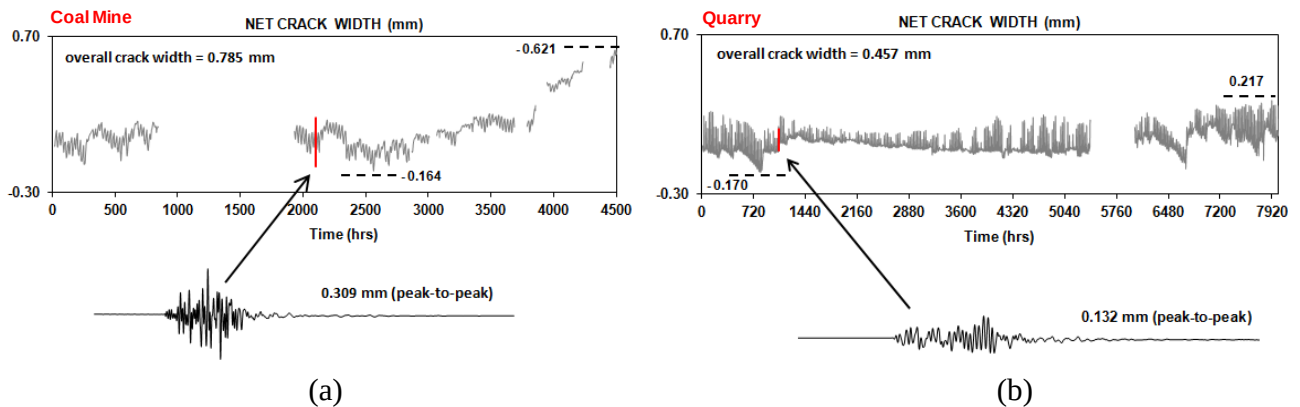


Figure 11: Dynamic vs. static crack response, for the coal mine (a) and quarry (b).

Conclusions

The comparisons presented, showing the influences of ground vibrations generated by blasting in hard and soft rock on the structures movements, indicate the strain levels induced in the walls by the same amplitudes of ground motions are greater in hard rock geology. This difference can be attributed to the higher powder factors used in quarrying, resulting in more energy released during the blast and by the local geology characteristics. The shape of the structure is also an important factor, contributing for the higher response of it.

It was also observed that the influence of weather variations on movement of an existing crack over time is much greater than during the short duration influence of seismic waves resulting from blasting. Finally, it is concluded that blasting cannot be considered a major source of damages to structures because the daily variations in temperature and humidity exert greater influence on the cyclical deformations of materials used in civil construction. Both the deformations calculated as the crack width variations and expansion of materials during blasting events did not significantly exceed the fatigue limits of the wall materials. It is the constant extensions and contractions of the materials comprising the wall induced by daily variations of temperature and humidity that, over time, result in the stress and fatigue of the materials.

This study also shows that the Brazilian Standard limits for safe blasting in urban areas (ABNT NBR 9653, 2005) is conservative and vibration levels below the proposed limits cannot possibly result in damages to structures.

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